

Graded Lab Activity (5 Marks)

LLM-Powered Autonomous Agents

Follow the instructions below to complete the task and submit your work for grading.

Instructions:

Initial Setup: Starter code is provided to set up the lab environment.

Sequential Steps: Follow the steps carefully to set up the LLM and the agent in sequence.

Observation & Enhancement: Run the agent and observe its outputs.

Modify or update the provided tool by enhancing its prompt or functionality so that your agent performs a meaningful task.

Submission

Ensure your final code demonstrates a working autonomous agent with at least one customized tool.

```
1 %%capture
2 !pip install langchain==0.2.5 langchain-community==0.2.5
   #sentence-transformers==3.0.1
```

Setting Up LLM

```
1 !pip install llama-cpp-python==0.2.78 --extra-index-url https://abetlen.github.io/llama-cpp-python/whl/cu124
```

```
Looking in indexes: https://pypi.org/simple, https://abetlen.github.io/llama-cpp-python/whl/cu124
Requirement already satisfied: llama-cpp-python==0.2.78 in /usr/local/lib/python3.12/dist-packages (0.2.78)
Requirement already satisfied: typing-extensions>=4.5.0 in /usr/local/lib/python3.12/dist-packages (from llama-cpp-python==0.2.78) (4.5.0)
Requirement already satisfied: numpy>=1.20.0 in /usr/local/lib/python3.12/dist-packages (from llama-cpp-python==0.2.78) (1.24.0)
Requirement already satisfied: diskcache>=5.6.1 in /usr/local/lib/python3.12/dist-packages (from llama-cpp-python==0.2.78) (5.6.1)
Requirement already satisfied: Jinja2>=2.11.3 in /usr/local/lib/python3.12/dist-packages (from llama-cpp-python==0.2.78) (3.1.2)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2>=2.11.3->llama-cpp-python==0.2.78) (2.1.5)
```

```
1 !wget https://huggingface.co/microsoft/Phi-3-mini-4k-instruct-gguf/resolve/main/Phi-3-mini-4k-instruct-gguf.2
```

```
--2025-11-17 12:14:34-- https://huggingface.co/microsoft/Phi-3-mini-4k-instruct-gguf/resolve/main/Phi-3-mini-4k-instruct-gguf.2
Resolving huggingface.co (huggingface.co)... 108.157.142.53, 108.157.142.55, 108.157.142.50, ...
Connecting to huggingface.co (huggingface.co)|108.157.142.53|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://cas-bridge.xetub.hf.co/xet-bridge-us/662698108f7573e6a6478546/df220524a4e4a750fe1c325e41f09ff69137f38b52c
--2025-11-17 12:14:34-- https://cas-bridge.xetub.hf.co/xet-bridge-us/662698108f7573e6a6478546/df220524a4e4a750fe1c325e41f09ff69137f38b52c
Resolving cas-bridge.xetub.hf.co (cas-bridge.xetub.hf.co)... 18.173.166.85, 18.173.166.82, 18.173.166.74, ...
Connecting to cas-bridge.xetub.hf.co (cas-bridge.xetub.hf.co)|18.173.166.85|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2393231072 (2.2G)
Saving to: 'Phi-3-mini-4k-instruct-q4.gguf.2'

Phi-3-mini-4k-instr 100%[=====] 2.23G 169MB/s in 26s

2025-11-17 12:15:00 (87.5 MB/s) - 'Phi-3-mini-4k-instruct-q4.gguf.2' saved [2393231072/2393231072]
```

```
1 from langchain.agents import initialize_agent, Tool, AgentType
```

```
1 from langchain.llms import LlamaCpp
```

Loading LLM

Load LLM of your choice.

```
1 # Load LLM Local for example Phi-3 Mini 4K gguf Model
2 llm = LlamaCpp(
3     model_path="Phi-3-mini-4k-instruct-q4.gguf",
4     n_gpu_layers=-1, # full GPU
5     max_tokens=500,
6     n_ctx=2048,
```

```

7     temperature=0.1,
8     verbose=False
9 )

```

Define Tools

One of the strengths of LangChain is the ability to build a **list of tools** that can then be used by the LLM.

To do this, we have to give a name, explain what the function to be performed is, and provide a description.

This way, the LLM is informed about what tool it can use when it has to do an internet search:

For more tools explore the following langchain tools:

<https://docs.langchain.com/oss/python/integrations/tools>

for enhancing or updating tools see below langchain tools use

<https://docs.langchain.com/oss/python/langchain/tools>

```

1 # example definition of a tool
2
3 def add_numbers(q):
4     nums = [int(x) for x in q.split() if x.isdigit()]
5     return f"Sum = {sum(nums)}"
6
7 # Use of tool
8
9 tools = [
10     Tool(
11         name="Addition Tool",
12         func=add_numbers,
13         description="Useful for adding integers in a query. Input should contain numbers."
14     )
15 ]

```

```

1 # for more than one tools use the following
2 '''
3 tools.append(
4     Tool(
5         name="name of tool",
6         func=function call,
7         description="Useful tool to search ---.",
8     )
9 )
10 '''

```

Create template

the template will serve as instructions to LLM

```

1 from langchain.prompts import PromptTemplate
2
3 template = """Answer the following questions as best you can. You have access to the following t
4
5 {tools}
6
7 Use the following format:
8
9 Question: the input question you must answer
10 Thought: you should always think about what to do
11 Action: the action to take, should be one of [{tool_names}]
12 Action Input: the input to the action
13 Observation: the result of the action
14 ... (this Thought/Action/Action Input/Observation can repeat N times)
15 Thought: I now know the final answer
16 Final Answer: the final answer to the original input question
17
18 Question: {input}

```

```
19 {agent_scratchpad}""  
20 prompt = PromptTemplate.from_template(template)
```

▼ Create and Execute Agent

```
1 from langchain.agents import AgentExecutor, create_react_agent
```

```
1 agent = create_react_agent(  
2     llm=llm,  
3     tools=tools,  
4     prompt=prompt  
5 )
```

```
1 agent_executor = AgentExecutor(  
2     agent=agent,  
3     tools=tools,  
4     verbose=True,  
5     max_iterations=1,  
6     handle_parsing_errors=True  
7 )
```

```
1 query = "7 7"  
2  
3 response = agent_executor.invoke({"input": query})  
4 print(response["output"])
```

> Entering new AgentExecutor chain...

*</assistant/> Question: What is the sum of 7 and 7?
Thought: To find the sum, I will use the Addition Tool.
Action: Addition Tool
Action Input: 7 + 7Sum = 14*

> Finished chain.
Agent stopped due to iteration limit or time limit.

Your tasks

To submit your work you need to do following:

Either Update tool definition to make it a complete calculator for adding/subtracting or multiplying numbers

or add another tool/tools to enhance agent capabilities

after completing submit the work via the given link